Master of Computer Application – General
Semester – I

Sub: ADVANCED DATA SCIENCE

Topic:.FAKE CURRENCY DETECTION USING DEEP LEARNING

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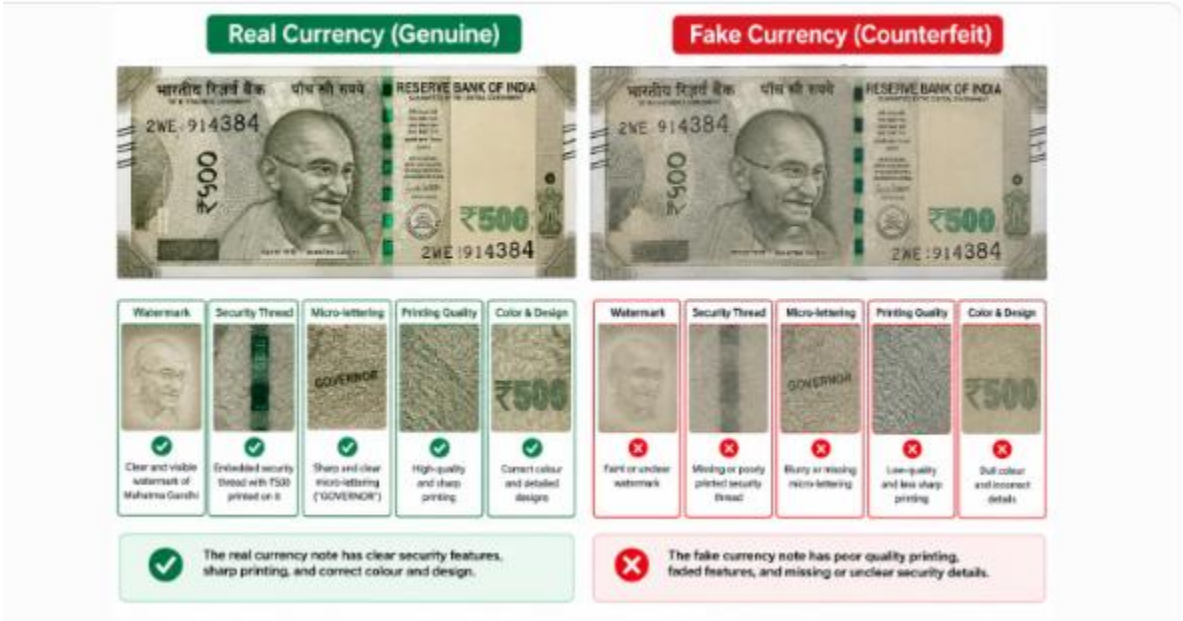
INTRODUCTION:

Fake currency detection is an important application of machine learning and deep learning that helps identify counterfeit currency notes and distinguish them from genuine notes. Counterfeit currency is a serious issue that can cause financial losses to individuals, businesses, and banks. Traditional methods of detecting fake currency mainly depend on manual inspection of security features, such as watermarks, security threads, serial numbers, and paper quality. However, manual verification can be time-consuming and may lead to human errors.

The Fake Currency Detection System Using Machine Learning aims to automate the process of identifying potentially fake currency notes. The system uses a dataset containing genuine and counterfeit currency information or images to train a machine learning model. Algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), and Convolutional Neural Network (CNN) can be used depending on the type of dataset. The trained model analyzes the input features or currency images and predicts whether a note is likely to be real or fake.

The main objective of this project is to develop a simple, efficient, and user-friendly system that helps users check currency notes more conveniently. The system can reduce the effort required for manual inspection and provide quick predictions. Its performance can be evaluated using metrics such as accuracy, precision, recall, and F1-score.

In conclusion, fake currency detection using machine learning demonstrates how artificial intelligence can be applied to financial security. Although the system can assist in identifying suspicious currency notes, its predictions depend on the quality of the dataset and the trained model and should not replace official currency verification procedures.


Real Currency (Genuine)
Fake Currency (Counterfeit)
RESERVE BANK OF INDIA
Watermark
Security Thread
Micro-lettering
Printing Quality
Color & Design
GOVERNOR
The real currency note has clear security features, sharp printing, and correct colour and design.
The fake currency note has poor quality printing, faded features, and missing or unclear security details.

PROBLEM STATEMENT:

The circulation of counterfeit currency is a serious problem that causes financial losses to individuals, businesses, and banks. Identifying fake currency manually can be difficult and time-consuming because counterfeit notes may look similar to genuine currency notes.

The main problem is to develop a Fake Currency Detection System using Machine Learning that can analyze currency note features or images and predict whether a note is real or fake. The system uses a trained machine learning model to identify patterns and differences between genuine and counterfeit notes based on the available dataset.

The proposed system aims to improve the speed and accuracy of currency verification, reduce human errors, and provide a simple, user-friendly method for detecting potentially counterfeit currency.

Objectives:

- *To collect and preprocess a currency dataset.*
- *To train a machine learning or deep learning model using genuine and counterfeit currency data.*
- *To classify currency notes as real or fake based on the model's input features.*
- *To evaluate the model's performance using suitable metrics such as accuracy, precision, and recall.*
- *To develop a simple interface for users to test currency notes.*

Expected Outcome: The system should predict whether a currency note is likely to be genuine or counterfeit based on the data it has been trained on. Its predictions are not a substitute for official currency verification, and its reliability depends on the quality of the dataset and model.

EXISTING SYSTEM:

In the existing system, fake currency detection is mainly performed through manual verification and traditional currency-checking methods. People, shopkeepers, and bank employees examine currency notes by checking security features such as the watermark, security thread, micro-lettering, serial number, colour, and paper quality.

Some banks and businesses also use counterfeit currency detection machines to identify suspicious notes. However, manual checking requires experience and can be time-consuming, while basic detection machines may have limitations in identifying certain types of counterfeit notes.

Disadvantages of the Existing System

1. *Time-consuming: Manually checking each currency note takes time.*
2. *Human errors: People may fail to identify counterfeit notes due to a lack of experience.*
3. *Limited accuracy: Visual inspection ma*
4. *y not detect sophisticated counterfeit notes.*
5. *Cost: Special currency detection machines may be expensive.*
6. *Lack of automation: Manual verification does not automatically analyze currency images using machine learning.*

Conclusion: The existing system relies primarily on manual inspection and specialized detection devices. These limitations create a need for an automated fake currency detection system using machine learning to analyze currency features and help identify potentially counterfeit notes.

PROPOSED SYSTEM:

The proposed system aims to develop an automated Fake Currency Detection System using Machine Learning or Deep Learning to identify whether a currency note is genuine or counterfeit. The system analyzes currency images or extracted features from a dataset and uses a trained model to classify the note as real or fake.

First, the currency dataset is collected and preprocessed to remove errors and prepare the data for training. Next, a machine learning algorithm is trained using the available currency data. When a user uploads a currency image or enters the required features, the system processes the input and predicts whether the currency note is real or fake. The result is displayed through a simple, user-friendly interface.

Advantages of the Proposed System

1. *Automation: Reduces the need for manual currency verification.*
2. *Faster detection: Provides predictions quickly after processing the input.*
3. *Improved consistency: Applies the same trained model to each input.*
4. *User-friendly interface: Allows users to submit currency images or features easily.*
5. *Performance evaluation: Uses accuracy, precision, recall, and other metrics to assess the model.*
6. *Reduced human error: Helps users identify potentially counterfeit notes.*

Conclusion

The proposed system uses machine learning techniques to make fake currency detection faster and more convenient. Its effectiveness depends on the quality of the dataset, the algorithm, and the input provided. The result should be treated as a preliminary prediction rather than official confirmation that a currency note is genuine or counterfeit.

SYSTEM SPECIFIACTION:

The Fake Currency Detection System using Machine Learning is designed to identify whether a currency note is real or fake by analyzing its image or selected features. The system uses machine learning or deep learning techniques to classify currency notes and display the prediction to the user.

1. Hardware Requirements

Processor: Intel Core i3 or above

RAM: Minimum 4 GB (8 GB recommended)

Hard Disk: Minimum 10 GB of free space

Input Device: Keyboard and mouse

Camera/Scanner: To capture currency note images (optional)

Display: Standard monitor

2. Software Requirements

Operating System: Windows 10/11, Linux, or macOS

Programming Language: Python 3.x

Development Environment: Jupyter Notebook, Google Colab, or VS Code

Frontend: HTML, CSS, and JavaScript (if a web interface is used)

Backend: Flask (for connecting the web interface with Python)

Libraries:

NumPy – numerical calculations

Pandas – dataset handling

Matplotlib – data visualization

Seaborn – statistical visualization

OpenCV – image processing

Scikit-learn – machine learning algorithms

TensorFlow/Keras – deep learning model development (if used)

3. Functional Requirements

The system should accept currency images or relevant currency features as input.

It should preprocess the input data or image.

It should use a trained machine learning model to analyze the input.

It should display the predicted result as Real or Fake.

It should provide a simple and user-friendly interface.

It should display an appropriate message if the uploaded image is invalid.

4. Non-Functional Requirements

Accuracy: The model should provide reliable predictions on suitable test data.

Performance: The system should process inputs within a reasonable time.

Usability: The interface should be simple and easy to understand.

Reliability: The system should handle errors and invalid inputs properly.

Security: Uploaded images and user data should be handled safely.

INPUT CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Fake Currency Detection</title>
 <style>
  * { box-sizing: border-box; }

  body {
   margin: 0;
   font-family: Arial, sans-serif;
   background: #eef3f8;
   color: #172b4d;
  }

  header {
   background: #0b2545;
   color: white;
   padding: 22px;
   text-align: center;
  }

  .container {
   max-width: 650px;
   margin: 40px auto;
   padding: 28px;
   background: white;
   border-radius: 16px;
   box-shadow: 0 8px 28px #152b4b18;
  }

  h2 { text-align: center; }

  .upload {
   border: 2px dashed #8ca5bd;
   border-radius: 12px;
   padding: 25px;
   text-align: center;
   margin: 22px 0;
  }

  input { max-width: 100%; }

  img {
   display: none;
   max-width: 100%;
   max-height: 280px;
```

```
    margin: 18px auto;
    border-radius: 8px;
   }

   button {
    width: 100%;
    padding: 14px;
    background: #087f8c;
    color: white;
    border: 0;
    border-radius: 8px;
    font-size: 16px;
    cursor: pointer;
   }

   button:disabled {
    opacity: .6;
    cursor: not-allowed;
   }

   #result {
    margin-top: 20px;
    padding: 14px;
    background: #edf5f8;
    border-radius: 8px;
    text-align: center;
    overflow-wrap: anywhere;
   }

   .note {
    color: #5b677a;
    font-size: 13px;
    line-height: 1.6;
   }

   footer {
    text-align: center;
    padding: 20px;
    color: #5b677a;
   }
  </style>
 </head>
 <body>
  <header>
   <h1>NoteVerify</h1>
   <p>Fake Currency Detection System</p>
  </header>

  <main class="container">
   <h2>Bank Note Authentication</h2>
```

```
  <p>Upload a clear image of a currency note for analysis.</p>

  <div class="upload">
    <label for="currencyImage">Choose a banknote image</label>
    <br><br>
    <input
      type="file"
      id="currencyImage"
      accept="image/png,image/jpeg"
    >
    <img id="preview" alt="Selected banknote preview">
  </div>

  <button id="checkButton" type="button">
    Check Currency
  </button>

  <div id="result" role="status" aria-live="polite">
    Please upload an image to begin.
  </div>

  <p class="note">
    Disclaimer: This is an educational project. A prediction
    cannot guarantee authenticity. Please consult an authorized
    bank for verification.
  </p>
</main>

<footer>
  Fake Currency Detection | Machine Learning Project
</footer>

<script>
  const input = document.getElementById("currencyImage");
  const preview = document.getElementById("preview");
  const result = document.getElementById("result");
  const button = document.getElementById("checkButton");

  let selectedFile = null;

  input.addEventListener("change", () => {
    selectedFile = input.files[0] || null;
    result.textContent = "";

    if (!selectedFile) {
      preview.style.display = "none";
      return;
    }

    if (!["image/jpeg", "image/png"].includes(selectedFile.type)) {
```

```
    selectedFile = null;
    input.value = "";
    preview.style.display = "none";
    result.textContent = "Please select a JPG or PNG image.";
    return;
  }

  if (selectedFile.size > 5 * 1024 * 1024) {
    selectedFile = null;
    input.value = "";
    preview.style.display = "none";
    result.textContent = "Image must be smaller than 5 MB.";
    return;
  }

  preview.src = URL.createObjectURL(selectedFile);
  preview.style.display = "block";
  result.textContent = "Image selected. Ready for analysis.";
});

button.addEventListener("click", async () => {
  if (!selectedFile) {
    result.textContent = "Please select a banknote image first.";
    return;
  }

  const data = new FormData();
  data.append("image", selectedFile);

  button.disabled = true;
  result.textContent = "Analyzing image...";

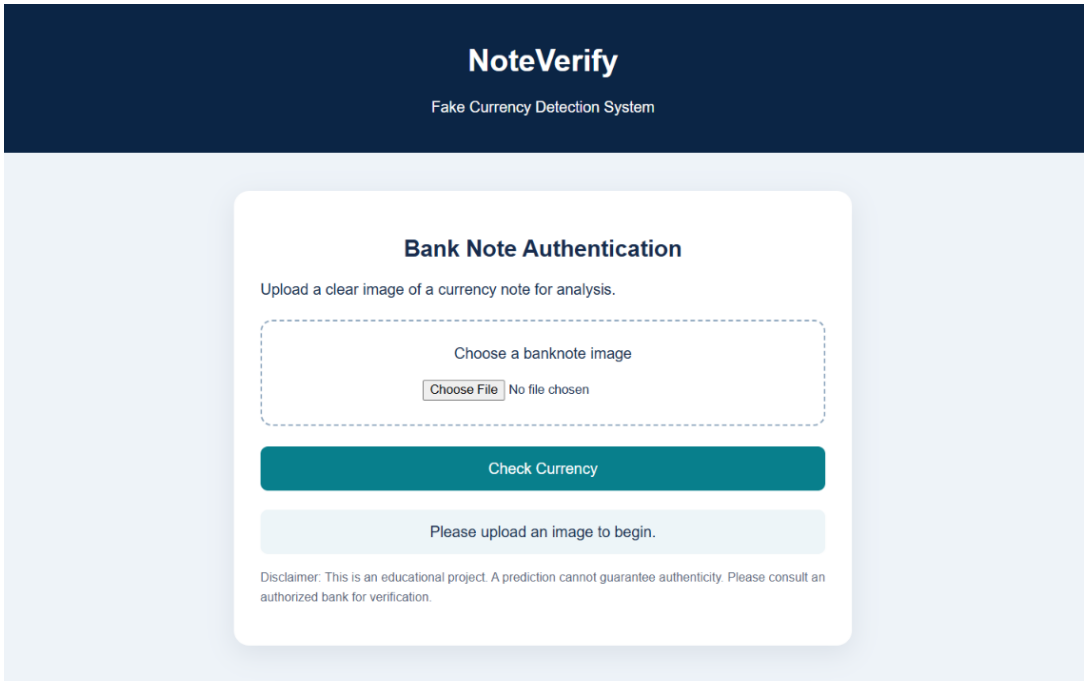
  try {
    const response = await fetch("/predict", {
      method: "POST",
      body: data
    });

    const output = await response.json();

    if (!response.ok) {
      throw new Error(output.error || "Analysis failed.");
    }

    result.textContent =
      "Model prediction: " + output.prediction +
      ". This result is not official verification.";
  } catch (error) {
    result.textContent =
      "Could not analyze the image. Start the Python backend " +
```

```
        "and connect a trained model. " + error.message;
      } finally {
        button.disabled = false;
      }
    });
  </script>
</body>
</html>
```



PYTHON CODE:

```
from flask import Flask, request, jsonify, render_template
from tensorflow.keras.models import load_model
from PIL import Image
import numpy as np
import io

app = Flask(__name__)

# Load your previously trained model
model = load_model("currency_model.keras")

IMG_SIZE = (224, 224)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/predict", methods=["POST"])
def predict():
    if "currencyImage" not in request.files:
        return jsonify({"error": "Please upload an image"}), 400

    file = request.files["currencyImage"]

    try:
        image = Image.open(io.BytesIO(file.read())).convert("RGB")
        image = image.resize(IMG_SIZE)

        # Adjust preprocessing to match your training code
        image_array = np.array(image, dtype=np.float32) / 255.0
        image_array = np.expand_dims(image_array, axis=0)

        prediction = model.predict(image_array, verbose=0)

        # Assumes a single sigmoid output:
        # 0 = Real, 1 = Fake
        score = float(np.asarray(prediction).reshape(-1)[0])

        label = "FAKE CURRENCY" if score >= 0.5 else "REAL CURRENCY"

        return jsonify({
            "prediction": label,
            "score": round(score, 4),
            "warning": "Model prediction only; not proof of authenticity."
        })

    except Exception as e:
        return jsonify({"error": str(e)}), 500

if __name__ == "__main__":
    app.run(debug=True)
```

CONCLUSION:

The Fake Currency Detection system using Machine Learning is designed to identify whether a currency note is real or fake by analyzing its image. This project uses image processing and machine learning or deep learning techniques to examine the features of currency notes and classify them based on a trained model.

The main objective of this project is to reduce the risk of counterfeit currency and help people identify suspicious banknotes more easily. The system allows users to upload an image of a currency note and receive a prediction through a simple user interface developed using HTML, CSS, JavaScript, and Python.

This project demonstrates how artificial intelligence and machine learning can be applied to real-world problems. The accuracy of the system depends on the quality of the dataset, the training process, and the model used. Although the system can assist in identifying suspicious currency, its predictions should not be considered official proof of authenticity. Future improvements may include using larger datasets, advanced deep learning algorithms, and additional security-feature analysis to improve reliability.

In conclusion, the Fake Currency Detection project provides a useful learning application of machine learning that can support counterfeit currency identification and promote awareness of currency security